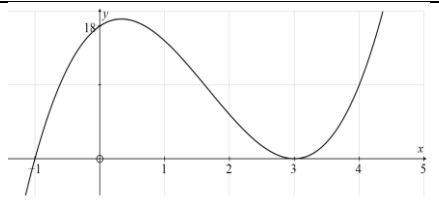
Question			Answer	Marks	AO	Guidance	
6	(a)		$y = \int (6x^2 - 20x + 6)\ dx = 2x^3 - 10x^2 + 6x + c$	M1	1.1a	Attempt to integrate	Do not allow if multiplied by x
			(2, 6) on curve so $6 = 2\times 2^3 - 10\times 2^2 + 6\times 2 + c \quad [c = 18]$	M1	1.1a		
			So $y = 2x^3 - 10x^2 + 6x + 18$	A1	1.1b	Complete equation must be seen	
				[3]			
6	(b)		$y = 2(x+1)(x^2 - 6x + 9)$ or $2(x^2 - 2x - 3)(x - 3)$	M1	2.1	o.e. Attempt to remove all brackets leading to a cubic expression. A quadratic factor must be seen.	
			So $y = 2x^3 - 10x^2 + 6x + 18$	E1	2.1	Correct expression found	
			Alternative method Division of $2x^3 - 10x^2 + 6x + 18$ by $(x+1)$ or $(x-3)$ and attempt to factorise quadratic factor	M1		Leading to a product of linear factors. A quadratic factor must be seen.	
			$2(x+1)(x-3)^2$	E1		Correct expression found	
			Alternative method [By Calculator] the roots of $y = 0$ are -1 and 3 [repeated] so $(x+1)$ and $(x-3)$ are factors	M1		Method using the factor theorem used with roots found by calculator or substitution in their (a)	
			$(x-3)$ is a repeated factor and the leading coefficient is 2 so $y = 2(x+1)(x-3)^2$	E1		Fully explained	
				[2]			

Question			Answer	Marks	AO	Guidance
6	(c)			**B1** **B1** **B1** **[3]**	**1.1b** **1.1b** **1.1b**	Correct shape *y*-intercept at 18 labelled Curve crosses *x*-axis at (–1, 0) and touches at (3, 0)

Question	Answer	Marks	AO	Guidance